

JAESEUNG KIM

Medical Imaging AI & Clinical ML

xmflak20@gmail.com

github.com/importjason

U.S. Citizen - no sponsorship required

EDUCATION

Yonsei University

Expected Feb 2027

B.S. Biomedical Engineering; Double Major in Electrical and Electronic Engineering

GPA 4.03/4.30 | Department Rank 1/49 | Percentile 97.9/100

RESEARCH EXPERIENCE

Clinical AI Research Intern, SNU iRAIL

2026 - Present

Task-driven clinical AI research in a hospital-facing environment, including clinical problem formulation, data review, model development, evaluation planning, and reproducible documentation.

Undergraduate Researcher, Advanced Image Processing Techniques Lab, Yonsei University

Dec 2023 - Feb 2026

Developed quantitative MRI prediction and neuroimaging pipelines; investigated acquisition and preprocessing choices; evaluated scanner and sequence generalization; and prepared two first-author manuscripts.

Undergraduate Researcher, Bio-Metamaterials Lab, Yonsei University

Jul 2022 - Aug 2022

Assisted with ellipsometer setup, biomaterial optical analysis, a tensile-test prototype, and a MATLAB user interface.

SELECTED RESEARCH

- **Acquisition-aware qT1 prediction**

Designed preprocessing and 3D regression pipelines, identified input-target resolution mismatch, and developed a minimal acquisition-aware target-alignment strategy.

- **Clinical imaging analysis and downstream validation**

Prepared qT1 maps, implemented voxel-wise models, and assessed whether predicted maps preserved population-level findings.

- **Computational MRI and NAMER reproduction**

Reproduced raw-k-space reconstruction and motion-estimation components and diagnosed failure modes before extension.

- **Task-driven clinical AI at SNU iRAIL**

Clinical problem formulation, data review, model development, evaluation design, and research documentation.

RESEARCH OUTPUTS

- Two first-author manuscripts in preparation.
- KSMBE Fall Conference 2025 - oral presentation and Excellent Paper Award.
- NEURO2026 - accepted conference presentation.
- ICMRI 2026 - quantitative neuroimaging presentation.

RESEARCH PRACTICE

- Quantitative MRI and acquisition modeling
- 3D medical-image learning and evaluation
- Raw-k-space reconstruction and motion modeling
- Spatial normalization and voxel-wise inference
- Domain generalization and harmonization
- Reproducible research pipelines and technical documentation

BEYOND MODELING

- Government R&D proposal development and technical work-package design.
- Cross-institutional and external-stakeholder coordination.
- Funded project execution, reporting requirements, and delivery timelines.
- Presentation, research documentation, and handoff-ready workflow development.

HONORS AND SCHOLARSHIPS

- Excellent Paper Award, Korean Society of Medical & Biological Engineering, 2025.
- National Science & Engineering Scholarship.
- Han Kyung Beom Scholarship.
- Yonsei University Academic Excellence Awards and six semesters of merit-based scholarship support.
- Programming and entrepreneurship awards.
- Military Training Excellence Award.